

Computer Science and Engineering Department
Motilal Nehru National Institute of Technology Allahabad
B.Tech IV Semester
Operating System Lab (CSN14401)
Assignment 1

1. Practice the following Linux commands: man, ls, cd, pwd, mkdir, rmdir, cat, less, head, tail, cp, mv, rm, touch, echo, ping, chmod, chown, exit, grep, diff, ps, top, kill, sudo, shutdown, vim, history, whoami, whatis, wc, more, cal, logout, I/O redirection commands (piping), etc.
2. Use the commands to show the following information of your system:
 - (i) CPU information
 - (ii) Memory information
3. Write C programs to simulate the following Linux commands:
 - (i) cd
 - (ii) ls
 - (iii) mkdir
 - (iv) grep
4. Write a C program which usage the concept of command line arguments.
5. Write a C program to calculate the execution time taken by insertion sort, selection sort and bubble sort to sort the registration numbers of students of your class. You can use the appropriate data structures and functions from time.h header file.